

1. Course number and name

CHE 455 Nanoscale Systems in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

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4. Text book, title, author, publisher, year

"Introduction to Nanoscience", S.M. LINDSAY. ISBN-10: 0199544212

a. Other supplemental materials

- a) Introduction to Nanomaterials and Devices by Omar Manasreh.
- b) Introduction to Nanoscience and Nanotechnology, Chris Binns
- c) Nanomaterials and Nano chemistry, by Bréchnignac, C., Houdy, P., Lahmani, M

5. Specific course information

a. Brief description of the content of the course (catalog description)

Basic principles associated with nanoscience and nanotechnology; fabrication and synthesis, size dependent properties, characterization; applications of materials at nanometer length scales; emphasis on recent technological breakthroughs.

b. Prerequisites or co-requisites

CHE 230; and Graduate or advanced undergraduate standing.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Explain the history of nanotechnology.
- Understand the nanoscale paradigm in terms of properties at the nanoscale dimension.
- Demonstrate a working knowledge of nanoscience/nanotechnology principles and applications
- Apply key concepts in materials science, chemistry, physics, biology, and engineering to the field of nanotechnology.
- Problem solving in homework/exams and by giving a presentation to the class.
- The student will be able to gain the ability to communicate effectively.

b. Explicitly indicate which of the student outcomes are addressed by the course.

	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(3) an ability to communicate effectively with a range of audiences.			x	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			x	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				x
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				x
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. **Brief list of topics to be covered**

Basic concepts and modern methods in major developments of nanotechnology and nanomaterials will be introduced. Tools and principles relevant to nanoscale materials will also be introduced. Current and future nanotechnology applications in engineering, materials, physics, chemistry, electronics, and energy will be discussed.

- Physics and Chemistry of Nanomaterials
- Application of Quantum Mechanics to Nanomaterials
- Electronic Structure of Nanomaterials
- Characterization Tools
- Nano-manufacturing
- Applications
 - Nanomaterials in engineering, physics, chemistry
 - Nanomaterial Examples – Energy
 - Nanomaterial Examples – Medicine
 - Nanomaterial Examples – Environmental applications
 - Nanomaterial Examples – Electronics
- Implications, Ethics, & Safety of Nanomaterials